

## BURNS, OR, USA

WMO: 726830

Lat: 43.595N

Lon: 118.956W

Elev: 4140

StdP: 12.63

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94185

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |             |             |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|-------------|-------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |             | 99%         |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>12     | (b)<br>-2.5 | (c)<br>4.1 | (d)<br>-7.3                   | (e)<br>4.3 | (f)<br>-0.7 | (g)<br>-1.0 | (h)<br>6.0 | (i)<br>6.6 | (j)<br>24.5           | (k)<br>39.2 | (l)<br>21.0 | (m)<br>35.1 | (n)<br>2.5            | (o)<br>260 | (p)<br>0.626 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>38.1            | (c)<br>93.5     | (d)<br>59.9 | (e)<br>90.8 | (f)<br>59.4 | (g)<br>87.8 | (h)<br>58.4 | (i)<br>63.2         | (j)<br>85.3 | (k)<br>61.0 | (l)<br>85.0 | (m)<br>59.6 | (n)<br>83.7 | (o)<br>7.7           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.6                            | 76.8 | 69.9 | 52.3 | 67.8 | 66.2 | 49.7 | 61.6 | 64.6 | 30.6          | 84.9 | 28.8 | 84.7 | 27.7 | 83.6 | 87.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |            |            |            |            |            |            |            |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|------------|------------|------------|------------|------------|------------|------------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |            | n=10 years |            | n=20 years |            | n=50 years |            |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max        | Min        | Max        | Min        | Max        | Min        | Max        | DB | WB |
| (a) 21.8          | (b) 18.7 | (c) 16.5 | (e) -11.5                  | (f) 98.9 | (g) 7.0            | (h) 3.1  | (i) -16.5                                          | (j) 101.1  | (k) -20.6  | (l) 102.9  | (m) -24.5  | (n) 104.6  | (o) -29.6  | (p) 106.9  |    |    |
| (4) 21.8          | (5) 18.7 | (6) 16.5 | (7) -11.5                  | (8) 98.9 | (9) 7.0            | (10) 3.1 | (11) -16.5                                         | (12) 101.1 | (13) -20.6 | (14) 102.9 | (15) -24.5 | (16) 104.6 | (17) -29.6 | (18) 106.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan      | Feb      | Mar      | Apr      | May      | Jun      | Jul      | Aug      | Sep      | Oct      | Nov      | Dec      |  |  |
|------------------------------------------------------------------------|-------------|--------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--|--|
|                                                                        |             | (d)    | (e)      | (f)      | (g)      | (h)      | (i)      | (j)      | (k)      | (l)      | (m)      | (n)      | (o)      | (p)      |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.2   | 26.0     | 29.8     | 37.8     | 43.1     | 51.6     | 59.0     | 68.3     | 65.8     | 56.8     | 44.6     | 34.0     | 25.2     |  |  |
|                                                                        | DBStd       | 16.34  | 9.25     | 7.87     | 6.62     | 6.85     | 7.50     | 7.36     | 5.82     | 5.74     | 7.37     | 7.30     | 8.51     | 9.31     |  |  |
|                                                                        | HDD50       | 3473   | 744      | 566      | 378      | 226      | 72       | 14       | 0        | 1        | 23       | 198      | 481      | 770      |  |  |
|                                                                        | HDD65       | 7469   | 1209     | 986      | 842      | 657      | 419      | 205      | 34       | 60       | 258      | 634      | 930      | 1235     |  |  |
|                                                                        | CDD50       | 1739   | 0        | 0        | 2        | 18       | 121      | 283      | 567      | 490      | 227      | 30       | 1        | 0        |  |  |
|                                                                        | CDD65       | 258    | 0        | 0        | 0        | 0        | 3        | 24       | 136      | 84       | 11       | 0        | 0        | 0        |  |  |
|                                                                        | CDH74       | 7404   | 0        | 0        | 0        | 13       | 192      | 851      | 3087     | 2404     | 808      | 49       | 0        | 0        |  |  |
|                                                                        | CDH80       | 3240   | 0        | 0        | 0        | 0        | 40       | 288      | 1534     | 1097     | 273      | 8        | 0        | 0        |  |  |
|                                                                        | WSAvg       | 6.5    | 4.9      | 6.0      | 7.4      | 8.3      | 7.9      | 7.5      | 6.9      | 6.6      | 6.3      | 6.0      | 5.4      | 5.1      |  |  |
|                                                                        | PrecAvg     | 11.7   | 1.9      | 1.1      | 1.1      | 0.7      | 0.9      | 0.7      | 0.3      | 0.6      | 0.5      | 0.7      | 1.5      | 1.8      |  |  |
| (15) Precipitation                                                     | PrecMax     | 16.0   | 5.7      | 2.1      | 2.4      | 2.1      | 2.3      | 1.9      | 1.1      | 2.0      | 1.2      | 3.7      | 3.2      | 5.5      |  |  |
|                                                                        | PrecMin     | 7.4    | 0.7      | 0.2      | 0.1      | 0.1      | 0.0      | 0.0      | 0.0      | 0.0      | 0.0      | 0.0      | 0.1      | 0.0      |  |  |
|                                                                        | PrecStd     | 2.2    | 1.1      | 0.6      | 0.8      | 0.6      | 0.7      | 0.6      | 0.3      | 0.7      | 0.4      | 0.8      | 0.8      | 1.1      |  |  |
|                                                                        | PrecStd     | 2.2    | 1.1      | 0.6      | 0.8      | 0.6      | 0.7      | 0.6      | 0.3      | 0.7      | 0.4      | 0.8      | 0.8      | 1.1      |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 48.6     | 56.2     | 67.0     | 75.8     | 84.6     | 92.6     | 98.9     | 96.0     | 90.8     | 79.8     | 63.5     | 49.9     |  |  |
|                                                                        |             | MCWB   | 41.3     | 44.7     | 47.1     | 51.8     | 57.0     | 60.7     | 62.7     | 60.8     | 57.4     | 53.6     | 47.6     | 43.3     |  |  |
|                                                                        | 2%          | DB     | 43.7     | 49.1     | 61.1     | 70.7     | 79.4     | 87.1     | 94.5     | 92.6     | 86.8     | 73.2     | 57.5     | 44.6     |  |  |
|                                                                        |             | MCWB   | 38.1     | 41.1     | 45.0     | 50.3     | 55.5     | 58.7     | 60.2     | 59.9     | 56.2     | 51.2     | 45.7     | 39.7     |  |  |
|                                                                        | 5%          | DB     | 40.3     | 44.6     | 56.3     | 64.8     | 74.6     | 83.0     | 91.9     | 90.0     | 83.1     | 67.8     | 52.2     | 41.2     |  |  |
|                                                                        |             | MCWB   | 36.0     | 38.3     | 43.2     | 47.4     | 53.5     | 57.1     | 59.8     | 58.9     | 55.3     | 48.8     | 43.5     | 37.1     |  |  |
|                                                                        | 10%         | DB     | 37.5     | 41.0     | 51.6     | 59.5     | 69.8     | 78.9     | 89.0     | 86.7     | 78.8     | 62.4     | 47.9     | 38.0     |  |  |
|                                                                        |             | MCWB   | 34.5     | 36.1     | 41.4     | 45.5     | 51.4     | 55.4     | 59.1     | 57.6     | 54.2     | 46.7     | 41.0     | 35.0     |  |  |
|                                                                        | 0.4%        | WB     | 43.0     | 45.7     | 49.5     | 53.4     | 59.5     | 64.0     | 69.9     | 69.4     | 63.3     | 55.3     | 50.3     | 43.9     |  |  |
|                                                                        |             | MCDB   | 46.4     | 54.7     | 62.7     | 71.6     | 78.5     | 86.9     | 87.5     | 90.0     | 77.3     | 73.9     | 59.5     | 47.9     |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 38.8     | 41.9     | 46.5     | 51.0     | 57.1     | 60.5     | 63.8     | 61.7     | 58.8     | 52.6     | 47.1     | 40.7     |  |  |
|                                                                        |             | MCDB   | 42.4     | 47.9     | 57.5     | 67.2     | 74.7     | 82.7     | 86.8     | 86.2     | 79.7     | 69.8     | 54.7     | 44.3     |  |  |
|                                                                        | 5%          | WB     | 36.6     | 39.0     | 44.4     | 48.6     | 54.9     | 58.4     | 61.6     | 59.8     | 56.8     | 50.1     | 44.4     | 37.6     |  |  |
|                                                                        |             | MCDB   | 39.6     | 43.7     | 54.5     | 62.8     | 71.5     | 79.4     | 86.3     | 85.5     | 79.6     | 65.0     | 51.6     | 40.2     |  |  |
|                                                                        | 10%         | WB     | 34.7     | 36.6     | 42.2     | 46.1     | 52.7     | 56.5     | 60.2     | 58.4     | 54.7     | 47.8     | 41.8     | 35.2     |  |  |
|                                                                        |             | MCDB   | 37.2     | 40.5     | 50.9     | 57.9     | 67.2     | 75.8     | 84.9     | 83.5     | 77.2     | 60.4     | 47.3     | 37.7     |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.5     | 19.3     | 23.2     | 26.5     | 28.6     | 32.3     | 38.1     | 39.1     | 38.1     | 32.8     | 23.6     | 18.9     |  |  |
|                                                                        |             | MCDBR  | 19.9     | 22.2     | 32.4     | 37.5     | 37.7     | 39.2     | 41.9     | 43.1     | 44.5     | 42.0     | 30.3     | 19.4     |  |  |
|                                                                        |             | MCWBR  | 15.0     | 15.0     | 18.6     | 19.7     | 17.3     | 16.9     | 16.4     | 18.2     | 20.5     | 23.2     | 19.7     | 14.9     |  |  |
|                                                                        | 5% WB       | MCDBR  | 17.6     | 20.6     | 29.0     | 33.7     | 33.6     | 36.1     | 38.5     | 39.9     | 40.1     | 36.7     | 27.6     | 16.9     |  |  |
|                                                                        |             | MCWBR  | 13.7     | 14.4     | 17.3     | 18.1     | 15.7     | 16.2     | 16.9     | 17.9     | 19.4     | 20.8     | 18.4     | 13.4     |  |  |
|                                                                        |             |        | (d) 18.5 | (e) 19.3 | (f) 23.2 | (g) 26.5 | (h) 28.6 | (i) 32.3 | (j) 38.1 | (k) 39.1 | (l) 38.1 | (m) 32.8 | (n) 23.6 | (o) 18.9 |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.260    | 0.267    | 0.275    | 0.295    | 0.308    | 0.308    | 0.315    | 0.311    | 0.294    | 0.285    | 0.264    | 0.259    |  |  |
|                                                                        | taud        |        | 2.560    | 2.595    | 2.601    | 2.534    | 2.516    | 2.542    | 2.532    | 2.528    | 2.562    | 2.590    | 2.596    | 2.562    |  |  |
|                                                                        | Ebn at Noon |        | 285      | 301      | 311      | 309      | 305      | 305      | 301      | 300      | 299      | 290      | 281      | 275      |  |  |
|                                                                        | Edh at Noon |        | 21       | 23       | 27       | 31       | 33       | 32       | 32       | 31       | 28       | 23       | 20       | 19       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 519      | 823      | 1222     | 1702     | 2038     | 2371     | 2465     | 2133     | 1647     | 1078     | 623      | 418      |  |  |
|                                                                        | RadStd      |        | 82       | 101      | 115      | 138      | 152      | 123      | 92       | 101      | 92       | 84       | 56       | 69       |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | -3.29   | -3.11  | N/A     | -0.76 | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (13 neighbors) | N/A   | N/A     | -2.56  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page